

Quiz (Carolina Reaper)

- Q1.** What is the slope of a line parallel to $y = 2x + 3$?
(A) $m = 1$
(B) $m = 2$
(C) $m = 3$
(D) $m = 4$
(E) $m = 5$
- Q2.** A shipping company has a delivery route with a slope of -2 . What is the slope of a parallel route?
(A) $m = -1$
(B) $m = -2$
(C) $m = -3$
(D) $m = 1$
(E) $m = 2$
- Q3.** The equation of a line is $y = 3x - 2$. What is the equation of a parallel line passing through $(1, 4)$?
(A) $y = 3x + 1$
(B) $y = 3x - 1$
(C) $y = 2x + 2$
(D) $y = 4x - 3$
(E) $y = 3x + 3$
- Q4.** A traffic flow model has two parallel roads with equations $y = 2x + 1$ and $y = 2x - 3$. What is the distance between them?
(A) 2 units
(B) 4 units
(C) 6 units
(D) 8 units
(E) 10 units
- Q5.** A logistics company has a route with a slope of $1/2$. What is the slope of a perpendicular route?
(A) $m = -2$
(B) $m = -1$
(C) $m = 1$
(D) $m = 2$
(E) $m = 3$
- Q6.** The equation of a line is $y = -x + 4$. What is the equation of a parallel line with y -intercept 2?
(A) $y = -x + 2$
(B) $y = -x - 2$
(C) $y = x + 2$
(D) $y = x - 2$
(E) $y = -2x + 2$
- Q7.** A delivery truck is traveling on a road with a slope of $3/4$. What is the slope of a parallel road?
(A) $m = 1/4$
(B) $m = 3/4$
(C) $m = 1$
(D) $m = 3$
(E) $m = 4$
- Q8.** The equations of two parallel lines are $y = 2x + 1$ and $y = 2x - 2$. What is the distance between the y -intercepts?
(A) 1 unit
(B) 2 units
(C) 3 units
(D) 4 units
(E) 5 units

Open Ended Questions (FRQ)

- Q9.** A transportation company has two parallel routes with equations $y = x + 2$ and $y = x - 1$. Find the distance between the routes and the point of intersection of the routes with a line perpendicular to them with equation $y = -x + 1$.
- Q10.** A shipping company has a delivery route with a slope of $-3/2$ and y -intercept 4. Find the equation of a parallel route with y -intercept 1 and the distance between the two routes.

Chef's Tips